Space–time focusing: breakdown of the slowly varying envelope approximation in the self-focusing of femtosecond pulses

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Numerical analysis of the self-focusing of femtosecond optical pulses reveals that the usually invoked slowly varying envelope approximation breaks down long before the temporal structure reaches the time scale of an optical cycle. This breakdown leads to a dramatic departure from the recently predicted symmetric development of the self-focusing pulse. By properly including the linear space–time focusing properties of short pulses, it is found that the trailing half of the self-focusing pulse is greatly enhanced and culminates in the formation of a trailing shock edge.

Self-focusing of optical pulses has been a topic of research for many years.[1–14] A thorough understanding has been achieved with steady-state (cw) self-focusing theory[1] and the quasi-cw moving-focus model[2] for nanosecond pulses. Recently there has been much research devoted to the understanding of the self-focusing of femtosecond pulses.[3–10] Anomalous observed continua spectra generated in gases have been connected with self-focusing.[3,4] High intensities readily available with short pulses have allowed self-focusing to be utilized as a mode-locking mechanism.[5] For the self-focusing of femtosecond pulses it has been shown that one must include the effects of group-velocity dispersion (GVD). In particular, for normal GVD it was found that a femtosecond pulse breaks up symmetrically into two pulses before the culmination of self-focusing.[6,7] For anomalous GVD, space–time self-collapse, instability, and self-trapped propagation have been predicted.[8–10] The analysis in this previous research accounted for GVD in the slowly varying envelope approximation[6–10] (SVEA). With this approximation one finds that propagation is described by a nonlinear Schrödinger equation (NLSE) with three transverse dimensions: two spatial dimensions and time. Because of its symmetry, the three-dimensional NLSE results in symmetric development of the self-focusing pulse. However, Bor[11] has recently shown that for short pulses, linear focusing by a lens results in temporal distortion owing to variation of the group delay of off-axis rays.

In this Letter it is shown that the three-dimensional NLSE derived in the SVEA is not adequate for describing the self-focusing of femtosecond pulses and that the breakdown of this approximation occurs for pulses much longer than an optical cycle. By analyzing the space–time nature of short-pulse focusing, it is found that the variation of the group velocity of off-axis rays causes an asymmetric development of the self-focusing pulse into a temporal structure with a greatly enhanced trailing portion.

The dependence of the slowly varying field $\mathscr{E}(z,x,y,t) = \mathrm{Re}\{E(z,x,y,t)\exp[-i(\omega_0 t - k_0 z)]\}$, propagating along z in a dispersive medium with cubic nonlinearity, can, to lowest order, be considered as governed by the NLSE[6–10]:

$$\frac{\partial E}{\partial z} + i[k^{(2)}/2]\frac{\partial^2 E}{\partial t^2} - \frac{i}{2k_0}\left(\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2}\right) = in_2(\omega_0/c)|E|^2 E, \quad (1)$$

where $k_0 \equiv n_0(\omega_0/c), k^{(2)} = \partial^2 k/\partial\omega^2|_{\omega_0}$ is positive for normal GVD, and a retarded time frame traveling at $v_g = \partial\omega/\partial k|_{\omega_0}$ is assumed. This equation assumes the slowly varying envelope and paraxial-ray approximations and an instantaneous nonlinearity, and the intensity dependence of the group velocity, the shock term[12] $(2n_2/c)\partial(|E|^2 E)/\partial t$, has been neglected.

To understand the approximations used in obtaining Eq. (1), consider the linear propagation equation obtained for the angular and temporal spectrum $\tilde{E}$ defined by $E(z,x,y,t) = \int\int\int \tilde{E}(z,k_x,k_y,\Omega)\exp[i(k_x x + k_y y - \Omega t)]\mathrm{d}k_x \mathrm{d}k_y \mathrm{d}\Omega$. On substitution of this equation into Eq. (1) and setting n_2 equal to zero, one obtains the correspondence $\partial/\partial x,y,t \longleftrightarrow ik_x, ik_y, -i\Omega$, and the linear propagation equation for the spectrum

$$\frac{\partial \tilde{E}}{\partial z} = i\left[\frac{k^{(2)}\Omega^2}{2} - \frac{k_x{}^2 + k_y{}^2}{2k_0}\right]\tilde{E} \equiv i\gamma\tilde{E}, \quad (2)$$

and therefore $\tilde{E}(z) = \tilde{E}(0)\exp(i\gamma z)$. Recalling the definition of E and the change to the retarded time frame in Eq. (1), one finds from Eq. (2) that $k_z = \gamma + k_0 + \Omega/v_g$. If, rather than starting from Eq. (1), one starts from the full-wave equation for linear propagation in a dispersive medium, it is found that a wave traveling in the $+z$ hemisphere has the dispersion relation

$$k_z = \left[k(\Omega)^2 - k_x{}^2 - k_y{}^2\right]^{1/2} \equiv \left[k(\Omega)^2 - k_T{}^2\right]^{1/2}, \quad (3)$$

where $k(\Omega)$ may be approximated by the series $k(\Omega) \cong k_0 + \Omega/v_g + k^{(2)}\Omega^2/2$. If the paraxial approximation

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is applied, for $k_T << k(\Omega)$, one obtains

$$k_z \cong k_0 + \Omega/v_g + k^{(2)}\Omega^2/2 - k_T^2/2k(\Omega). \quad (4)$$

If it is now assumed that the pulse envelope is slowly varying enough such that $k(\Omega)$ in the denominator of relation (4) can be approximated by k_0, then one obtains the previous relation found for k_z in connection with Eq. (2) and hence the SVEA (NLSE) of Eq. (1). However, it is this slowly varying approximation that dramatically fails in the description of self-focusing of short pulses. The physical result of this approximation amounts to ignoring any change in the z-projected group velocity of an off-axis ray owing to the variation of its tilt angle. This is a poor approximation for the *linear or nonlinear* propagation of a short pulse that has a significant angular spectrum. The correct physics describing the group velocity of these rays shall be referred to as *space–time focusing*. For this effect to be investigated, $k(\Omega)$ in the denominator in relation (4) can be expanded to $O(\Omega)$ to obtain

$$k_z - k_0 - \Omega/v_g \cong \frac{k^{(2)}\Omega^2}{2} - \frac{k_T^2}{2k_0}\left(1 - \frac{\Omega}{k_0 v_g}\right). \quad (5)$$

The predominant new term beyond those present in Eq. (2), $(k_T^2\Omega/2k_0^2 v_g)$, now is seen to be the variation of the group velocity of a tilted ray projected onto the z axis, v_{gz}. This follows because the coefficient of Ω in relation (5) is v_{gz}^{-1} and can be confirmed directly from geometrical considerations. That is, $v_{gz}^{-1} = v_g^{-1}/\cos\theta \cong v_g^{-1}(1 + \theta^2/2)$ in the paraxial approximation, where $\theta \cong k_T/k_0$ is the ray angle with the z axis, and thus $v_{gz}^{-1} \cong v_g^{-1}(1 + k_T^2/2k_0^2)$.

Integration of the self-focusing problem with the inclusion of space–time focusing is performed with the use of the split-step method, in which the linear and nonlinear propagation are calculated in small alternating segments.[13] Because the linear segment is calculated in the frequency domain, space–time focusing as described by Eq. (3) is easily implemented exactly without using the approximations of relations (4) and (5), and the measured value of $k(\Omega)$ is used rather than the Taylor series expansion. This is then the generalization of Ref. 13 to include fully material dispersion without assuming a limited bandwidth or invoking the paraxial approximation. It is of note that one could alternatively just include the lowest-order space–time focusing term in a modified NLSE description by adding $(-1/2k_0^2 v_g)\partial(\partial^2E/\partial x^2 + \partial^2E/\partial y^2)/\partial t$ to the left-hand side of Eq. (1). Calculations show that, in using this modified NLSE, one obtains substantially the same results as those presented here. It should also be noted that the space–time focusing treatment still ignores coupling to backward-traveling waves, higher-order nonlinearities, and the shock term.[12] These effects are likely to be significant as the time (and/or spatial) structure approaches the optical period (and/or diffraction limit)[14] and may be comparable with the effects of space–time focusing.

The effect of the inclusion of space–time focusing is found by comparison with the integration of the NLSE, Eq. (1). As a first example, consider self-focusing in a normally dispersive medium. Parameters typical of recent experiments in continuum generation are used. It is assumed that $\lambda = 600$ nm and that the medium is 40 atm of Xe [$k^{(2)} = 5 \times 10^{-3}$ ps^2/m, and $k(\Omega)$ values are taken from Ref. 15 and $n_2 = 3.6 \times 10^{-17}$ cm^2/W (Ref. 4)]. The input pulse is taken to have a sech^2 temporal intensity profile with a FWHM of 50 fs and a collimated transverse Gaussian profile of FWHM 80 μm. The peak input power of 29 MW is taken to be 1.9 times the critical power for the catastrophic collapse of a cw Gaussian beam,[1] $P_{\text{crit}} = (1.22)^2\pi\lambda_0^2/32n_0n_2 =$ 15 MW.

Figure 1 shows the axial intensity obtained by the numerical integration of Eq. (1) in the vicinity of the focus. As found previously,[7] the pulse symmetrically splits into two pulses that then continue to self-focus. Figure 2 shows the integration of the same problem, but now with the use of the non-SVEA, which incorporates space–time focusing based on the exact implementation of Eq. (3). One sees the dramatic shift of energy to longer times, which leads to a strongly asymmetric breakup of the input pulse into a weak leading pulse (suppressed to the point of appearing only as a pedestal) and a strong self-focusing trailing pulse. In addition, the trailing pulse appears to be forming a shock on its trailing edge. The newly observed behavior is understood qualitatively in terms of the variation of the axially projected group velocity with ray angle. That is, as the input pulse approaches the nonlinear focus, its angular spectrum increases. Once the wide-angle components become significant, the effect of the increase in v_{gz}^{-1} becomes apparent in that a significant fraction of the pulse energy is distributed in these tilted rays and hence is shifted to longer times. Thus, although the input pulse splits into two parts as is also found in Fig. 1, the shift of energy owing to space–time focusing promotes the self-focusing of the trailing portion of the pulse, whereas the leading portion is suppressed.

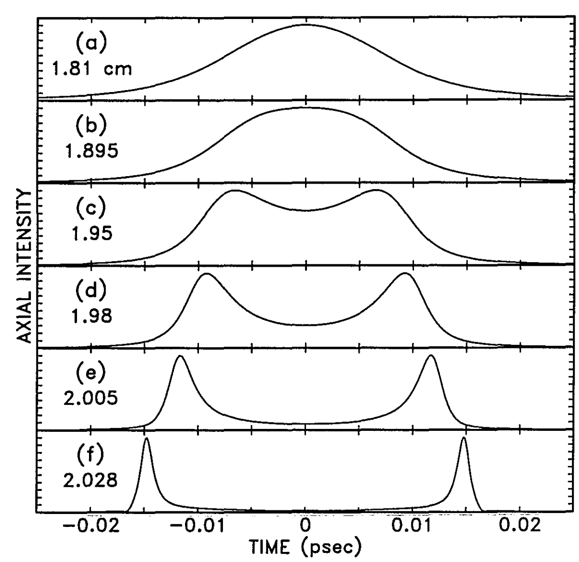

Fig. 1. Intensity on axis versus time of a self-focusing pulse in a normally dispersive medium at the propagation distances indicated, calculated with the use of the NLSE [Eq. (1)]. The peak intensities relative to the input are (a) 30, (b) 62, (c) 97, (d) 150, (e) 320, and (f) 600.

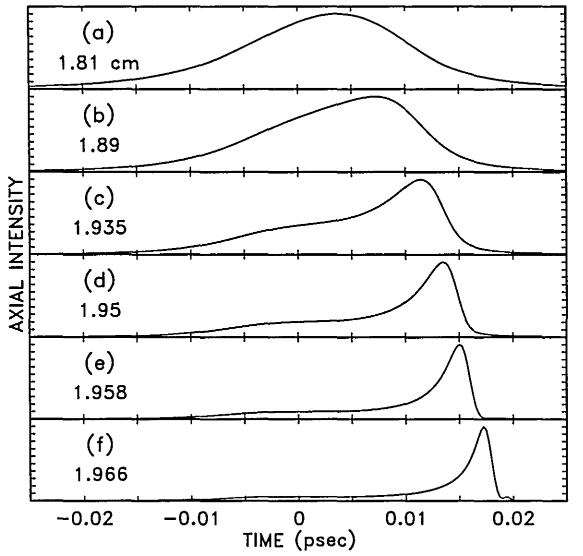

Fig. 2. Same as in Fig. 1, except calculated in a non-SVEA by including space–time focusing based on Eq. (3). The peak intensities relative to the input are (a) 29, (b) 61, (c) 140, (d) 270, (e) 540, and (f) 990.

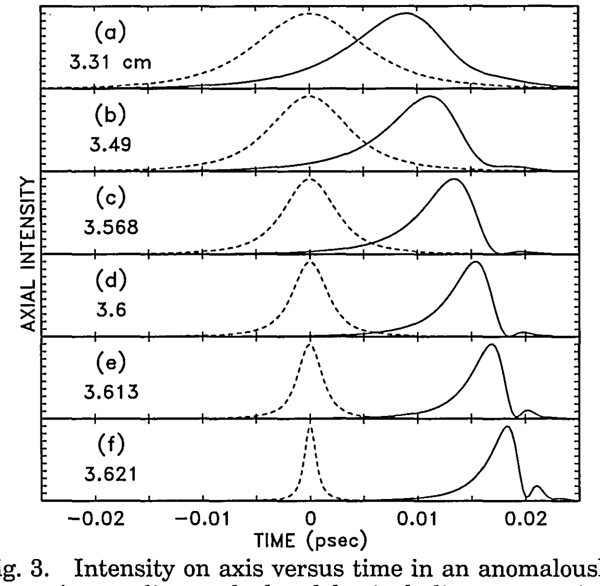

Fig. 3. Intensity on axis versus time in an anomalously dispersive medium calculated by including space–time focusing in a non-SVEA based on Eq. (3) (solid curves), at the propagation distances indicated, and by the NLSE [Eq. (1), dashed curves] for which the propagation distances are less by 1.0 mm. The respective peak intensities relative to the input for the non-SVEA and the NLSE calculations are (a) 9, 7; (b) 20, 15; (c) 55, 35; (d) 150, 80; (e) 330, 170; and (f) 640, 740.

The physical effect described here also must be considered for self-focusing in the anomalous dispersion regime, where pulse self-collapse has been predicted.[8,9] For example, consider the self-focusing of the same input pulse used previously (50-fs duration and 80-μm diameter) in fused silica ($n_2 = 3.2 \times 10^{-16}\ \mathrm{cm^2/W}$) for anomalous GVD [$\lambda = 1.5\ \mu$m and $k^{(2)} = -0.022\ \mathrm{ps^2/m}$]. Here also Eq. (3) is implemented exactly by using the measured value of $k(\Omega)$ for fused silica[16] to include space–time focusing and perform a non-slowly varying, nonparaxial calculation. The input-pulse peak power is taken to be $6.8\ \mathrm{MW} = 0.95P_{\mathrm{crit}}$. The space–time focusing solution (solid curves) is compared with the integration of the NLSE of Eq. (1) (dashed curves) in Fig. 3. As before, the NLSE yields symmetrical temporal behavior. By using the non-slowly varying method, as the pulse approaches the focus, the shift to later times becomes apparent, and one sees that it develops an asymmetric shape. The rear edge of the pulse appears to be forming a shock edge, and behavior similar to optical wave breaking[17] occurs. It also should be noted that, depending on the temporal and spatial scale, the predicted self-trapped solutions[9] found for anomalous GVD may no longer exist or be significantly modified when space–time focusing is accounted for.

In conclusion, it has been shown that the SVEA (i.e., the NLSE) fails to describe accurately the self-focusing of femtosecond pulses in dispersive media. In both the normal and anomalous GVD regions, one finds that the self-focusing pulse develops a highly asymmetric pulse shape, where the trailing portion of the pulse is enhanced. This effect is a result of including space–time focusing: the reduced axially projected group velocity of wide-angle rays in the angular spectrum of the self-focusing pulse.

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